

RAZIEH GHAEDI

☎ 07960177520 ✉ rozaghaedi90@gmail.com 🔗 LinkedIn 🐙 GitHub 🎓 Google Scholar 📍 Sheffield, UK

Professional Summary

Motivated MSc Computer Scientist (Distinction) with a strong research background in deep learning, multimodal learning, and human-centered AI. Experienced in designing and evaluating data-driven models for cross-domain facial expression recognition, affective computing, and socially responsible AI applications. Demonstrated research output through peer-reviewed publications (Knowledge-Based Systems, ACML 2025, IEEE Access) and interdisciplinary collaboration at the University of Sheffield. Research interests encompass machine learning, deep learning, and AI applications, with particular focus on interpretable, robust, and socially responsible intelligent systems.

Education

MSc in Computer Science

Sep 2023 – Oct 2024

Manchester Metropolitan University

Manchester, UK

- **GPA:** Distinction (80/100)
- **Dissertation:** Deep Learning-Based Cross-Domain Facial Expression Recognition (Grade: 80/100)
- **Key Modules:** Advanced Object-Oriented Programming, Algorithms and Data Structures, Cloud and Enterprise Development, Information Systems and Databases, Advanced Computer Networks, AI Ethics and Governance

BSc in Information Technology Engineering

Sep 2009 – Feb 2014

- **Key Modules:** Data Structures, Algorithms Design, Advanced Programming, Operating Systems, Database Design, Artificial Intelligence, Digital Electronics, Software Engineering

Publications

- **Ghaedi, R.,** BabaAhmadi, A., Alam, N., Fan, X. (2025). *Graph-Attention Network with Adversarial Domain Alignment for Robust Cross-Domain Facial Expression Recognition*. Presented at **ACML 2025**. [Paper] | [Poster]
- Zarina, O., Saleh Pour, M. J., **Ghaedi, R.,** Shafei Khah, M. (2023). *Markov-Based Reliability Assessment for Distribution Systems Considering Failure Rates*. **IEEE Access**, 11, 10018–10031. DOI:10.1109/ACCESS.2023.10032136
- Shirdel, M., **Ghaedi, R.,** Fan, X. (2025). *SACA: Selective Attention-Based Clustering Algorithm*. Under review at Knowledge-Based Systems journal. [Paper]
- Vafadarnikjoo, A., **Ghaedi, R.,** Imran, N. (2025). *A Text Mining and Topic Modelling Analysis of UK Modern Slavery Act Statements to Assess Effectiveness of Grievance Disclosures*. Under review at the *International Journal of Computational Intelligence Systems*.
- **Ghaedi, R.,** Farhang, D., Vafadarnikjoo, A. (2025). *Tentative title: A Multi-Agent AI Framework for Forced Labour Risk Assessment in Global Supply Chains*. Manuscript in preparation (experiments completed; writing in progress). Target journal: *International Journal of Production Research* (Special Issue: The Agentic Supply Chain: Entering a New Era in AI-Driven Supply Chain Management).
- Vafadarnikjoo, A., **Ghaedi, R.** (2025). *A Machine Learning and Simulation Approach for Improving Supply Chain Resilience*. Book chapter under review. Target book: *Computational Intelligence Techniques for Sustainable Supply Chain Management*. [GitHub]

Research and Professional Experience

Research Associate

2024 – Present

University of Sheffield

Sheffield, UK

- Developed a **multi-agent AI system** for automated corporate ethics benchmarking, enabling rapid assessment of forced labour risks across global ICT supply chains through parallel data processing and predictive analytics.
- Applied NLP and topic modeling (**LDA, BERTopic**) to analyze UK Modern Slavery Act statements, evaluating grievance mechanism effectiveness through systematic text mining of corporate disclosures.
- Designed and implemented a **Decision Support System** using Best-Worst Method (BWM) and Spanning Tree Enumeration (STE) for complex decision-making in supply chain management contexts.
- Contributing to **multiple interdisciplinary publications** bridging AI, operations management, and social impact research (manuscripts under review and in preparation).

Teaching Assistant

2025 – Present

rTriibe (Education Recruitment Agency)

Sheffield, UK

- Assisted classroom teachers across primary and secondary schools with lesson delivery, classroom management, and one-to-one pupil support, fostering a safe and engaging learning environment.

Software Engineer

2014 – 2021

Novin Asan Tech Mana Company

Shiraz, Iran

- Developed and maintained Java-based enterprise applications with emphasis on backend logic and relational database integration
- Led the migration of legacy systems to modern frameworks, improving system maintainability and user experience
- Collaborated with cross-functional teams to troubleshoot, deploy, and document software modules

Selected Research Projects

Multimodal Personalized Depression Detection *[GitHub]*

Built a multimodal, personalized depression detection system that fuses voice, facial cues, and personal attributes to predict depression level using attention-based deep learning (MPDD Challenge @ ACM Multimedia 2025).

Corporate Ethics Automation System (Kaggle Competition) *[GitHub]*, *[Kaggle]*

Kaggle Agents Intensive hackathon entry: built a Google ADK automation tool to replicate KnowTheChain benchmark workflows in <60 seconds using PDF extraction, sentiment analysis, correlation analysis, and regression.

RepoForge AI: Agentic Test Generation for Public Python Repositories (Kaggle Project)

Built a multi-agent system (OrchestratorAgent plus specialised agents for ingestion, code analysis, API discovery, retrieval, test planning, and test writing) that scans public Git repositories and generates validated pytest suites. Combined RAG over ChromaDB-indexed code chunks with Gemini-based generation, plus safety checks (secret redaction, unsafe-pattern blocking) to produce reviewable, downloadable test artifacts.

Graph-Attention Network for Cross-Domain FER *[GitHub]*

Integrated ResNet-50 and Graph Attention Networks with adversarial domain alignment (MMD, CORAL, GRL). Achieved 74% mean accuracy across six datasets (RAF-DB, FER2013, CK+, JAFFE, ExpW, SFEW2.0).

LDA-Based Grievance Mechanism Analysis *[GitHub]*

Applied advanced LDA topic modeling to 2,957 grievance documents from UK Modern Slavery Act statements, identifying 18 optimal topics to analyze patterns in corporate disclosure practices.

Technical Skills

Programming & Scripting: Python (Advanced), Java (Advanced), C# (Intermediate)

Machine Learning & AI: Deep Learning, Graph Neural Networks (GNNs), Transfer Learning, Domain Adaptation, Multimodal Learning, Representation Learning

Data Analysis & Visualisation: Advanced Excel, JMP, Power BI, Tableau, SQL, Statistical Analysis, Exploratory Data Analysis (EDA)

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers

Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS (EC2, S3)

Tools & Development: Git, Jupyter, LaTeX, REST APIs

Academic Service

Peer Reviewer

2025 – Present

Discover Artificial Intelligence (Springer Nature)

IEEE Transactions on Consumer Electronics

Achievements & Certifications

GRE: 328 (Verbal 158, Quantitative 170, AWA 4.0)

The Art and Science of Uncertainty and the Future (Interdisciplinary Summer School)

Jul 2026

DL4SG (Data Lab for Social Good), Cardiff University

Cardiff, UK

- Completed hands-on workshops on decision quality and probabilistic reasoning, communicating and visualising uncertainty, and predictive inconsistency in algorithmic risk assessment tools.

- Engaged with editor-led panels on publishing uncertainty and futures research, and built connections with an international, interdisciplinary community of ECRs and PhD researchers.

IEEE Computer Society Member (No. 100476059)

References

Dr. Nashid Alam

MSc Supervisor, Lecturer, Manchester Metropolitan University

n.alam@mmu.ac.uk

Dr. Rahim Taheri

BSc Supervisor, Senior Lecturer, University of Portsmouth

rahim.taheri@port.ac.uk

Dr. Amin Vafadarnikjoo

Research Supervisor, Lecturer, University of Sheffield

a.vafadarnikjoo@sheffield.ac.uk